No.

Date.

Ex. 3.2 | Ch.3 | P81 | Chloe | 2025.7.21

解：(0,1)区间，$f(x)=x^{\nu}$ 要在希尔伯特空间，需满足 $\int_0^1 |x^{\nu}|^2 dx = \int_0^1 x^{2\nu} dx < +\infty$

(a) $\lim\limits_{\varepsilon\to 0^+} \int_\varepsilon^1 x^{2\nu} dx = \lim\limits_{\varepsilon\to 0^+} \left[\frac{x^{2\nu+1}}{2\nu+1}\right]\Big|_\varepsilon^1$

$2\nu+1>0$，$\nu>-\frac{1}{2}$ 时，$\lim\limits_{\varepsilon\to 0^+}\left(\frac{1}{2\nu+1}-\frac{\varepsilon^{2\nu+1}}{2\nu+1}\right)=\frac{1}{2\nu+1}$，收敛

$2\nu+1=0$，$\nu=-\frac{1}{2}$ 时，$\int_0^1 x^{2\nu} dx = \int_0^1 x^{-1} dx = \ln x\Big|_0^1 = +\infty$ 发散

$2\nu+1<0$，$\nu<-\frac{1}{2}$ 时，原式 $=\frac{1}{2\nu+1}-\lim\limits_{\varepsilon\to 0^+}\frac{\varepsilon^{2\nu+1}}{2\nu+1}=\frac{1}{2\nu+1}-\frac{+\infty}{2\nu+1}=-\infty$ 发散

故当 $\nu>-\frac{1}{2}$ 时，$f(x)=x^{\nu}$ 在希尔伯特空间

(b) $\nu=\frac{1}{2}$ 时，$f(x)=x^{\frac{1}{2}}$，$\int_0^1 |x^{\frac{1}{2}}|^2 dx = \int_0^1 x\, dx = \frac{x^2}{2}\Big|_0^1 = \frac{1}{2} < +\infty$，在

$\int_0^1 |x\cdot x^{\frac{1}{2}}|^2 dx = \int_0^1 x^3 dx = \frac{x^4}{4}\Big|_0^1 = \frac{1}{4} < +\infty$，在

$\frac{df(x)}{dx} = \frac{1}{2}x^{-\frac{1}{2}}$，$\int_0^1 \left(\frac{1}{2}x^{-\frac{1}{2}}\right)^2 dx = \frac{1}{4}\int_0^1 x^{-1} dx = \frac{1}{4}\lim\limits_{\varepsilon\to 0^+} \ln x\Big|_\varepsilon^1 = +\infty$，不在